Tehran University of Medical Sciences

School of Nutritional Sciences and Dietetics · Graduate Seminar

# Ultra-Processed Foods and Addictive-Like Eating Behaviors

## From Neurobiological Mechanisms to Community Nutrition Policy




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Course instructor: Dr. Pouraram · Autumn 2026

# Roadmap

01 **Ultra-processed foods**
What are they and how much do we eat?

02 **Food addiction**
Valid construct? How measured?

03 **Mechanisms**
Why might UPF act differently?

04 **Human evidence**
What do trials and cohorts show?

05 **The debate**
Is “addictive” the right word?

06 **Policy**
What works without settling the debate?

07 **Vulnerable groups and clinic**
Children, eating disorders, stigma

08 **Conclusions**
Take-home messages for Iran

**Guiding question:** must we settle whether food is “addictive” before acting on ultra-processed foods?

*Click a circle to jump to that section; each divider links back here.*

# Why this question, why now?

**58%**

of energy in US adults' diets comes from NOVA group 4 (NHANES 2017–18)

**54%**

of energy in UK adults; about 66% in adolescents aged 11–18

**13.4%**

median UPF share of energy among Tehran adults (TLGS)

**14%**

of adults meet YFAS "food addiction" criteria in pooled estimates (12% of children)

## The puzzle



Obesity is rising simultaneously across very different populations — individual choice alone cannot explain it (GBD 2019; NCD-RisC).



UPFs dominate food supplies in high-income countries and are expanding fast in middle-income countries.



Some products are engineered to be highly rewarding. Could an "addiction" lens explain compulsive overconsumption?

*Sources: Steele et al. 2023; Rauber et al. 2020; Moslehi et al. 2024; Gearhardt et al. 2023; GBD 2019 Risk Factors Collaborators 2020; NCD-RisC 2024; Monteiro et al. 2013; Baker et al. 2020.*

# Separating the claims: unequal evidence

| | CLAIM | EVIDENCE NEEDED | WHAT WE HAVE | STATUS |
|---|---|---|---|---|
| a | **UPF intake ↔ higher energy intake and adverse health** | Controlled feeding trials, large prospective cohorts, umbrella reviews | Hall 2019, Dicken 2025, Lane 2024, NutriNet-Santé, TLGS | RELATIVELY STRONG |
| b | **Some UPFs ↔ addictive-like eating** | Validated psychometrics, behavioral experiments, longitudinal designs | Schulte 2015, YFAS surveys, LaFata 2024 — mostly cross-sectional | LIMITED |
| c | **These behaviors are a valid "addiction"** | Discriminant validity vs BED, predictive validity, stability, specific treatment response | Gearhardt vs Hebebrand and Ziauddeen; absent from DSM-5-TR and ICD-11 | UNRESOLVED |
| d | **Environmental policies change purchases and products** | Quasi-experimental evaluations (interrupted time series, pre–post, purchase data) | Chile, Mexico, United Kingdom — intermediate outcomes | STRONG (INTERMEDIATE) |

**The rows are independent:** rejecting (c) does not invalidate (a) or (d) — so policy need not wait for the addiction debate to end.

# How the evidence was assembled

### 1 Search

PubMed/MEDLINE, Web of Science, Scopus, Google Scholar

### 2 Prioritize

Umbrella reviews, meta-analyses, RCTs, cohorts >10,000, Iranian studies

### 3 Weigh

Design, risk of bias, precision, directness, consistency — not recency

### 4 Synthesize

117 cited sources: peer-reviewed papers, laws and official reports

SEARCH TERMS

- ultra-processed food
- NOVA classification
- food addiction
- Yale Food Addiction Scale
- hyper-palatable food
- addictive-like eating
- front-of-pack labelling
- sugar-sweetened beverage tax

**A narrative review, not a systematic one:** no fully reproducible search, no pre-registered criteria, no dual screening, no PRISMA flow and no uniform risk-of-bias appraisal. Conclusions are a structured narrative synthesis.

# Two research lines converge




Addiction science  Policy  Processing trials

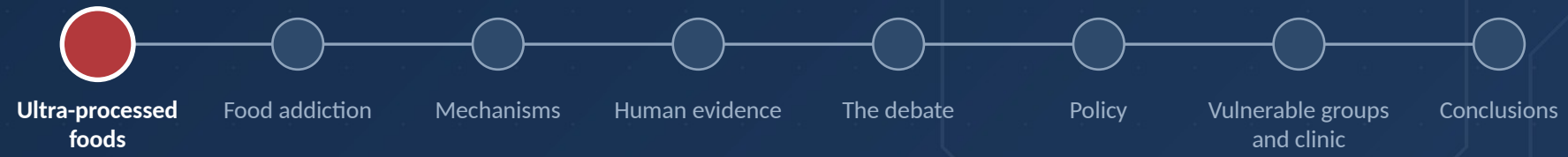
*Sources: Randolph 1956; Wang 2001; Avena 2008; Gearhardt 2009; Johnson and Kenny 2010; Ziauddeen 2012; Hebebrand 2014; Brazil MoH 2014; Taillie 2020; Scarborough 2020; Hall 2019; Gearhardt and Schulte 2021.*

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01

# Ultra-Processed Foods

*What are they — and how much of our diet do they make up?*



# NOVA: four groups by extent and purpose of processing

GROUP 1

## Unprocessed or minimally processed

Fruit, vegetables, legumes, eggs, fresh meat, milk, whole grains

GROUP 2

## Processed culinary ingredients

Oils, butter, salt, sugar, starch

GROUP 3

## Processed foods

Fresh bread, cheese, canned vegetables, smoked fish, pickles

GROUP 4

## Ultra-processed foods (UPF)

Soft drinks, packaged snacks, confectionery, sausages, ready meals

**Increasing extent and purpose of industrial processing**

**Spotting group 4:** an ingredient rarely used in home kitchens (high-fructose corn syrup, hydrogenated or interesterified oils, maltodextrin, protein isolates) or a cosmetic additive (flavors, colors, emulsifiers, sweeteners, thickeners).

*Sources: Monteiro et al. 2018; Monteiro et al. 2019.*

# What makes UPF different?

### High energy density

More energy per gram — higher intake, weaker satiety

### Disrupted food matrix

Texture and particle size change chewing, eating rate and nutrient availability

### Fat + refined carbohydrate

Rare in whole foods; a supra-additive effect on reward

### High glycemic load, low fiber

Rapid absorption; linked to hunger, craving and addiction-like behavior

### Soft texture, fast eating

Less chewing → faster eating → more energy consumed

### Sensory and technological additives

Flavor enhancers serve appeal; others serve texture or shelf life

### Less favorable nutrient profile

At diet level: more free sugars, saturated fat and sodium; less fiber

**9.9% → 15.4%**

energy from free sugars, lowest → highest UPF quintile (UK NDNS)

**Fiber: 8.4 → 6.9 g per 1000 kcal**

*Sources: Monteiro et al. 2019; Forde et al. 2020; Fardet 2024; Wang et al. 2026; Gearhardt and Schulte 2021; Robinson et al. 2014; Hall et al. 2019; Rauber et al. 2018.*

# NOVA under fire — and terms not to confuse

## Four critiques

**Ambiguous operational criteria**

Borderline foods are classified inconsistently

**Processing ≠ nutritional quality**

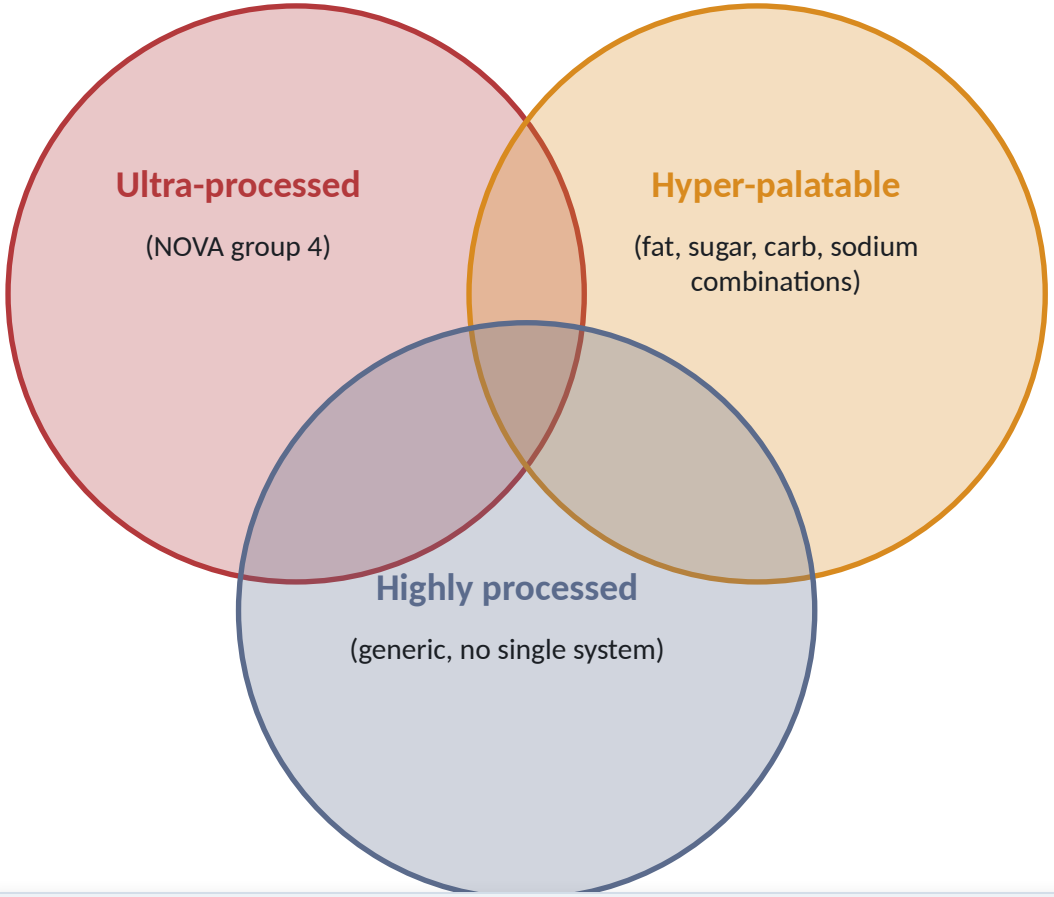
Group 4 members differ widely in nutrient profile

**Roles of processing blurred**

Policy must respect the group's heterogeneity

**Competing systems**

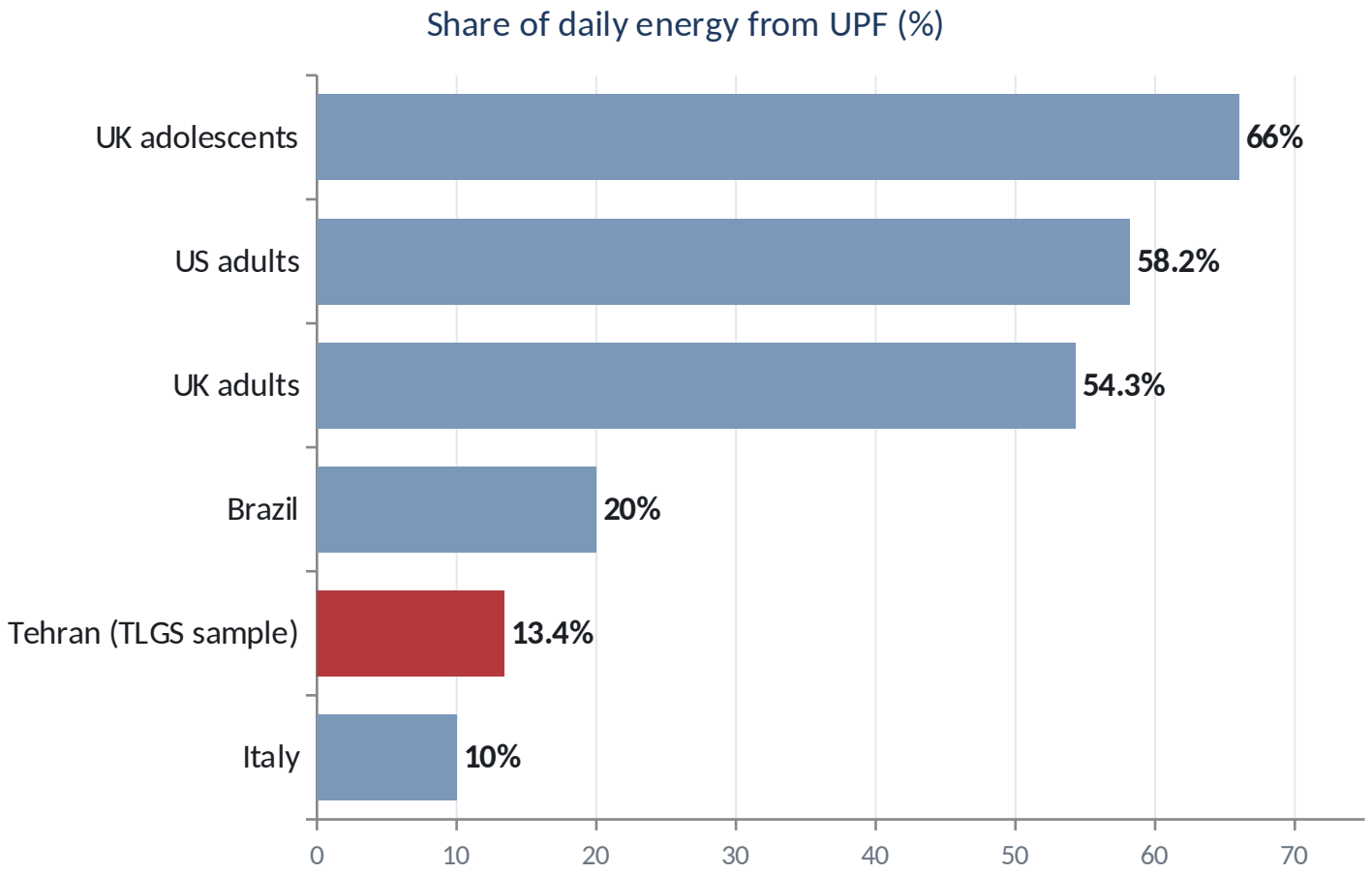
NOVA, IARC-EPIC, IFPRI and UNC give different estimates for the same diet




**Also distinct:** food addiction ≠ addictive eating ≠ loss-of-control eating

*Sources: Petrus et al. 2021; Gibney 2019; Moubarac et al. 2014; Astrup and Monteiro 2022; Fazzino et al. 2019.*

# How much UPF do we eat?




## Iran: fragmentary data

- TLGS (Tehran): median 13.4% of energy (~91 g/day)
- Highest vs lowest intake: prediabetes HR 1.28 (1.07–1.52); +10% UPF → HR 1.12
- No significant association with type 2 diabetes
- Other estimates: 8.5% (five cities) to 20.2% (n = 1,459 adults)
- Children aged 6 (n = 788): no link with overweight (OR 0.86)

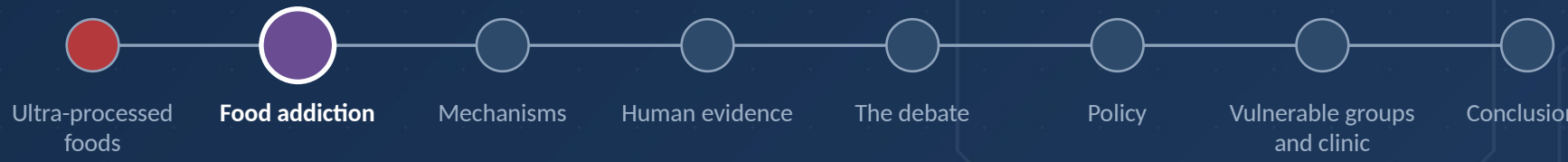
**Gap:** no regular national monitoring of UPF intake

*Descriptive comparison only (different years, ages, dietary methods, NOVA coding). Sources: Steele 2023; Rauber 2020; Marino 2021; Moslehi 2024; Haghighatdoost 2022, 2023; Asgari 2022.*

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# 02 Food Addiction

*Is it a valid construct — and how do we measure it?*



# What “food addiction” is — and is not

**Working definition:** eating with features resembling substance use disorders — intense craving, repeated failed attempts to cut down, time spent obtaining or eating food, continued eating despite harm, and sometimes withdrawal-like symptoms.

**DSM-5-TR (2022):** Not a recognized diagnosis

**ICD-11:** Not a diagnosis — “disorders due to addictive behaviours” cover only gambling and gaming

## Binge-eating disorder

**Official diagnosis**

Recurrent binges with loss of control; no compensatory behavior

## Bulimia nervosa

**Official diagnosis**

Recurrent binges plus inappropriate compensatory behaviors

## Food addiction

**Research construct**

Dependence-like relationship with food; binge episodes not required

*Sources: Gearhardt and Schulte 2021; Meule and Gearhardt 2014; American Psychiatric Association 2022; World Health Organization 2022; Hebebrand et al. 2014.*

# Translating DSM-5 substance-use criteria to eating

| DSM-5 criterion | Possible equivalent in eating | Evidence |
|---|---|---|
| Larger amounts or longer than intended | Eating more, or for longer, than intended | Empirical support |
| Persistent desire or failed efforts to cut down | Failed attempts to cut down or control intake | Empirical support |
| Much time obtaining, using, recovering | Time spent obtaining food, overeating, recovering | Conceivable |
| Craving | Intense craving for specific foods | Empirical support |
| Failure to fulfil major roles | Overeating impairs work, school or home roles | Conceivable |
| Use despite social problems | Overeating despite interpersonal problems | Conceivable |
| Activities given up | Activities reduced because of overeating | Conceivable |
| Hazardous use | Overeating in physically hazardous situations | Conceivable |
| Use despite physical or psychological harm | Overeating despite known harm | Empirical support |
| Tolerance | More food needed for the same effect | Conceivable |
| Withdrawal | Irritability or restlessness when abstaining | Conceivable; confounded by energy deficit |



## An asymmetric standard

No drug meets every criterion: caffeine lacks several, and tobacco has no intoxication syndrome.



## Lower thresholds, higher prevalence

DSM-IV needed ≥3 symptoms; DSM-5 “mild” needs ≥2. Direct transfer risks overestimating food addiction.

*Sources: Meule and Gearhardt 2014; American Psychiatric Association 2013.*

# Measuring it: the Yale Food Addiction Scale

**2009**

### YFAS

DSM-IV based · 25 items · 7 symptoms + distress · cut-off ≥3 symptoms + distress

**2013**

### YFAS-C

Adapted for children and adolescents

**2016**

### YFAS 2.0

DSM-5 based · 35 items · 11 symptoms · mild, moderate, severe

**2017**

### mYFAS 2.0

Short form · 13 items · 11 symptoms

## Persian validations

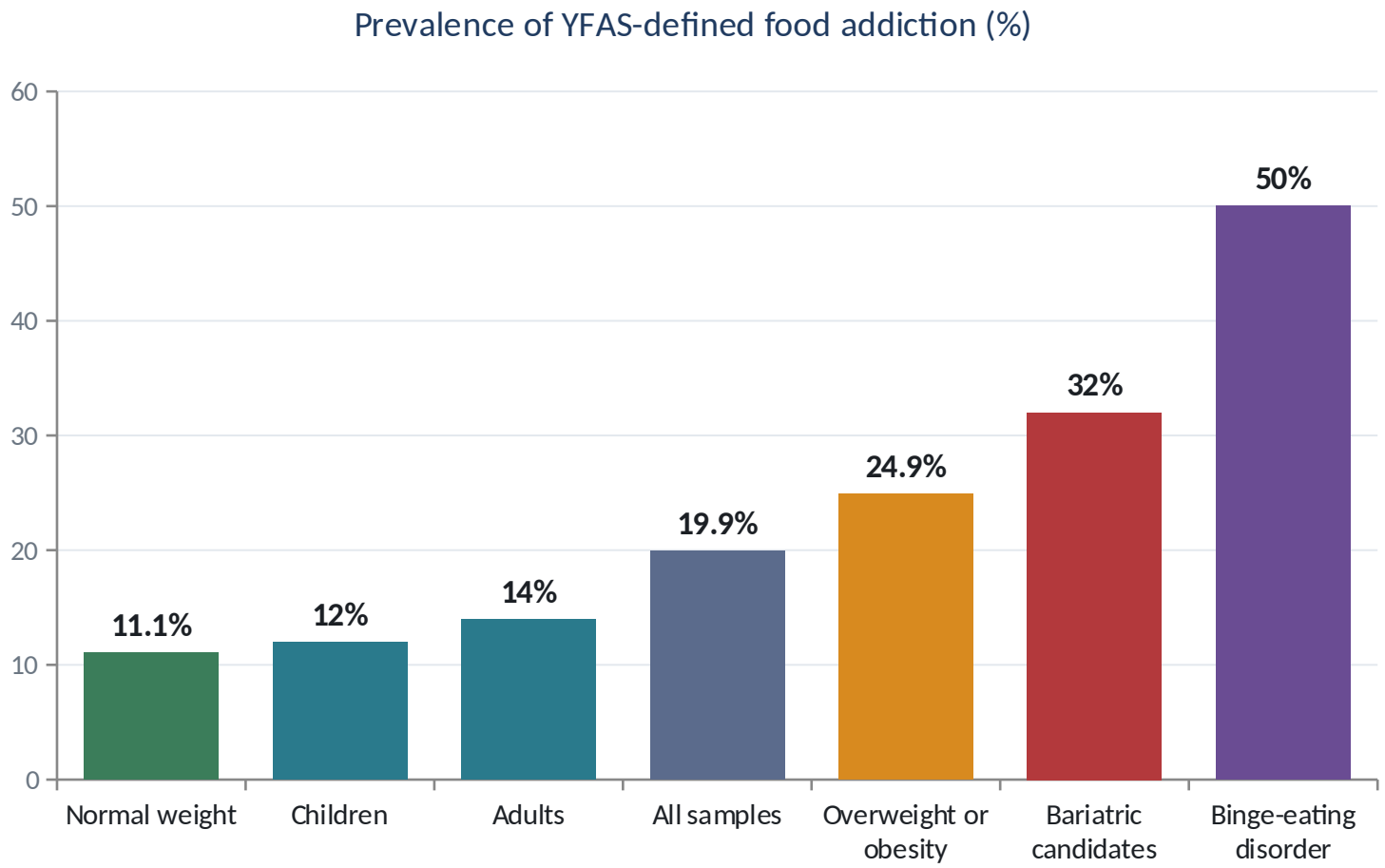
- **Ghanbari 2022:** YFAS 2.0, 451 students: single factor; r = 0.58–0.72 with food craving
- **Tajik-Esmaeeli 2024:** Bariatric candidates: α = 0.89, ICC = 0.88
- **Niroumand Sarvandani 2024:** mPYFAS 2.0, n = 9,606: α = 0.83; gender invariance

## The circularity problem

A tool built from substance-use criteria cannot, on its own, prove the construct exists. Does YFAS capture anything beyond binge eating, emotional eating, loss-of-control eating and restraint?
**→ Read prevalence as "YFAS-defined food addiction".**

*Sources: Gearhardt et al. 2009, 2013, 2016; Schulte and Gearhardt 2017; Ghanbari et al. 2022; Tajik-Esmaeeli et al. 2024; Niroumand Sarvandani et al. 2024; Ziauddeen and Fletcher 2013.*

# How common is YFAS-defined food addiction?




### Depends on who, which version, which cut-off

Pooled 19.9% (95% CI 16.3–24.0) across 25 studies; BED bar = “more than 50%”.

### A corrected meta-analysis

Meule (2024): bulimia estimates in Praxedes et al. clashed with primary studies (often >80–95%); a correction confirmed subgroup underestimation.

### 14% vs 13.9% is not equivalence

NESARC-III alcohol use disorder (13.9%) comes from diagnostic interviews; food addiction from self-report.

*Sources: Pursey et al. 2014; Gearhardt et al. 2023; Meule 2024; Praxedes et al. 2022; Grant et al. 2015, 2016.*

# Which foods are most linked to addictive-like eating?

## Most associated

*ultra-processed · high fat · high glycemic load*

1. Pizza
2. Chocolate
3. Potato chips
4. Cookies
5. Ice cream
6. French fries
7. Cheeseburger
8. Soda (non-diet)

## Least associated

*unprocessed or minimally processed*

1. Cucumber
2. Carrots
3. Beans (no sauce)
4. Apple
5. Brown rice
6. Broccoli
7. Banana
8. Water

## The abuse-potential analogy

**Dose ↑** more fat and refined carbohydrate per bite

**Speed ↑** faster absorption of refined carbohydrate

*An analogy, not pharmacokinetic equivalence: digesting a meal is not a drug reaching the brain.*

Forced-choice ranking of 35 foods (2015); predates NOVA and the term UPF.

*Sources: Schulte, Avena and Gearhardt 2015; Gearhardt and Schulte 2021; Gearhardt and DiFeliceantonio 2023; Hebebrand et al. 2014; Westwater et al. 2016.*

# Withdrawal: the weakest link

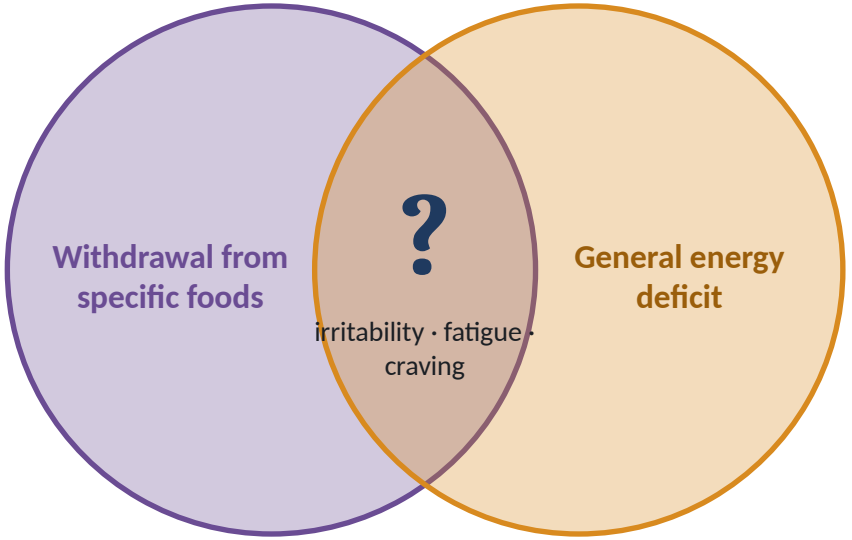
## What has been measured

**Highly Processed Food Withdrawal Scale (ProWS)**

- Irritability
- Fatigue
- Restlessness
- Craving

**Peak severity:** days 2–5 after cutting down

*Recent reviews still call this evidence preliminary.*




*If people cut sweets and calories together, irritability cannot be pinned on "sugar withdrawal".*

## Why the evidence is weak

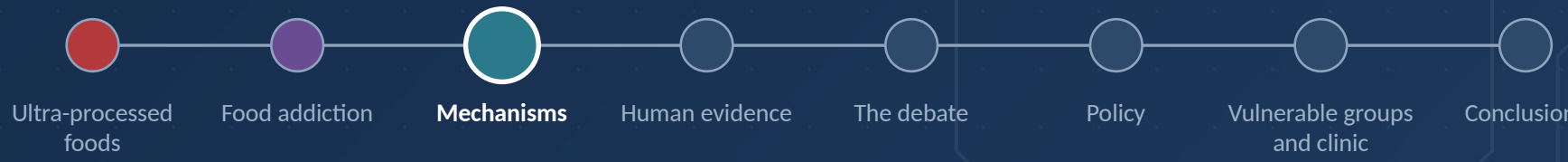
- 55% of patients with BED endorsed the YFAS withdrawal item — with no operational definition
- Retrospective self-report
- Expectancy in people who already believe they are "addicted"
- No objective biomarker
- Unclear time course; non-specific symptoms

*Sources: Schulte et al. 2018; LaFata et al. 2024; Gearhardt et al. 2012; Ziauddeen and Fletcher 2013; Meule and Kübler 2012; Westwater et al. 2016.*

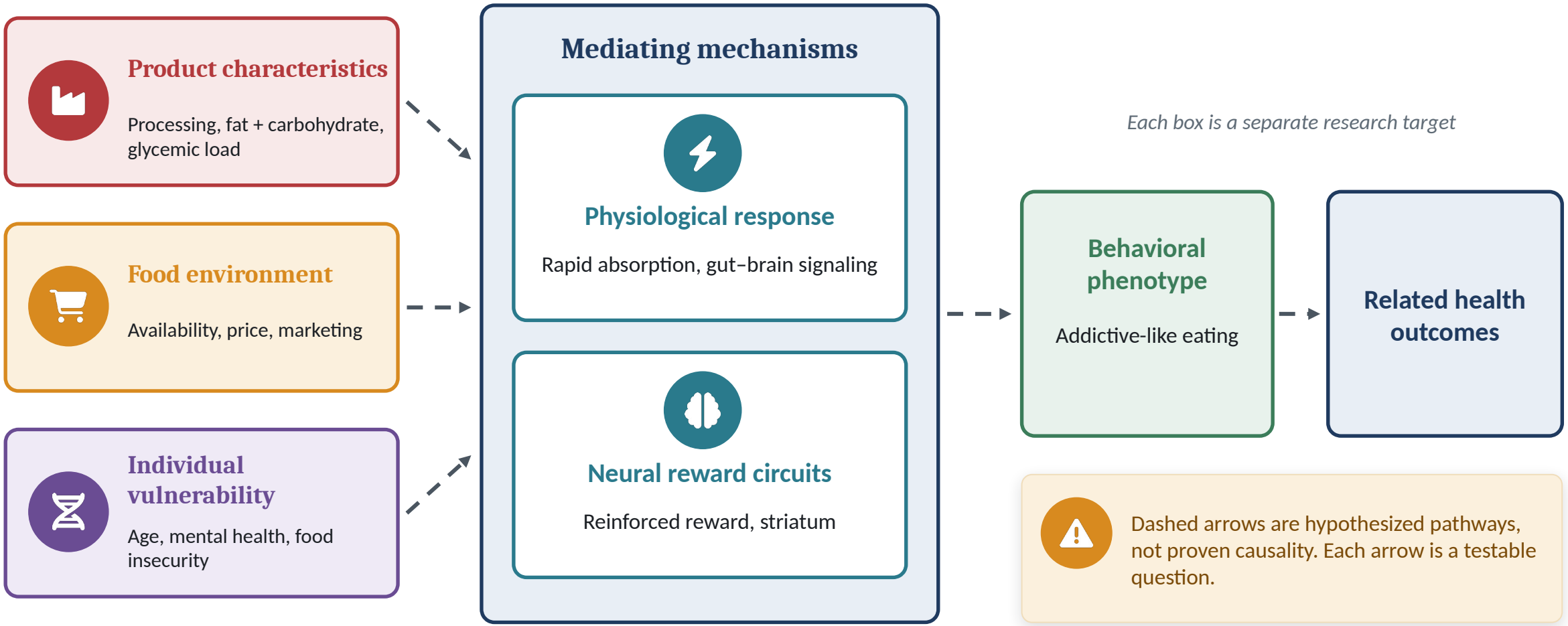
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# 03 Mechanisms

*Why might ultra-processed foods behave differently?*



# An integrated conceptual model




*Model compiled by the author from Schulte and Gearhardt; DiFeliceantonio et al. 2018; Volkow et al. 2013; Gearhardt et al. 2023; LaFata et al. 2024.*

# Delivery speed and the fat + carbohydrate combination

## Speed of delivery

In addiction science, faster delivery means stronger reinforcement. Processing may speed up refined-carbohydrate delivery — a hypothesis, not a proven mechanism.

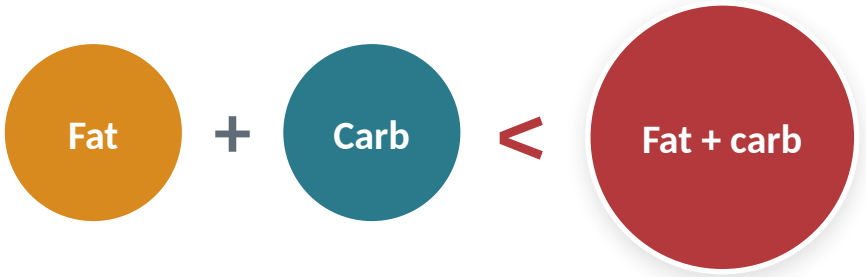
Haber et al. 1977

| | Whole apple | Fiber-free juice |
|---|---|---|
| **Eating speed** | slower | **much faster** |
| **Insulin rise** | smaller | **greater** |
| **Glucose afterwards** | steady | **rebound fall** |

*Juice is not necessarily NOVA group 4: the lesson concerns food structure.*

## Fat + carbohydrate: supra-additive

DiFeliceantonio et al. 2018




*more than the sum of its parts*

- Higher willingness to pay
- Greater response in reward-valuation regions, including the striatum
- Foods matched for familiarity, reported liking and calories

**Evolutionary hypothesis:** such rich combinations were rare in natural foods (breast milk is the cited example). Untested in humans — and a supra-additive effect is not an addictive effect.

*Sources: Schulte and Gearhardt; Haber et al. 1977; DiFeliceantonio et al. 2018; Small and DiFeliceantonio 2019.*

# Dopamine: three competing models

## Reward deficiency

**Reduced dopamine signaling drives eating more to reach the same reward.**

- **Wang, Volkow et al. 2001:** lower striatal D2 receptor availability in obesity (PET) — cross-sectional, so no causal direction
- **Johnson and Kenny 2010:** cafeteria-diet rats lost D2 availability and kept eating despite punishment — an animal model

**Predicts: blunted reward sensitivity**

## Incentive sensitization

**Repeated exposure sensitizes cue-triggered "wanting" — without more "liking".**

- **Robinson and Berridge:** motivation to obtain a reward dissociates from the pleasure of consuming it
- A person may want a food more without enjoying it more

**Predicts: heightened cue reactivity**

## Habit to compulsion

**Control shifts from goal-directed action to habit; dorsal striatal circuits take over.**

- **Everitt and Robbins:** behavior persists even after the reward is devalued
- Compulsion need not mean "needing more reward"

**Predicts: insensitivity to outcomes**

**Different predictions, not one model.** Citing only findings that fit a favored model artificially strengthens the construct (Ziauddeen, Farooqi and Fletcher 2012).

*Sources: Wang et al. 2001; Johnson and Kenny 2010; Robinson and Berridge 2008; Everitt and Robbins 2005; Ziauddeen et al. 2012; Volkow et al. 2013.*

# Opioids, endocannabinoids, genes and GLP-1 drugs

## Opioids: “liking”

- Opioids carry hedonic “liking”; dopamine carries “wanting”
- OPRM1 A118G and DRD2 Taq1A variants studied in binge-eating disorder (Davis et al. 2009)
- Association is not causation

## Genetics: weak support

- Candidate-gene findings (e.g., DRD2 Taq1A) replicate poorly (Border et al. 2019)
- Genome-wide approaches now preferred (Hatoum et al. 2023)
- No robust genetic overlap established

## Endocannabinoids: a cautionary tale

- CB1 activation boosts eating motivation and hedonics
- **Rimonabant** (CB1 antagonist): weight loss, but depression and suicidal ideation — withdrawn from the EU market (Christensen et al. 2007)

## GLP-1-based drugs

- **Semaglutide, n = 30, 12 weeks:** less food craving, better eating control (Blundell 2017)
- **Alcohol use disorder, n = 48:** fewer drinks per drinking day, less craving; drinking days unchanged (Hendershot 2025)
- **Animal dopamine data mixed;** tirzepatide is a dual GIP/GLP-1 agonist

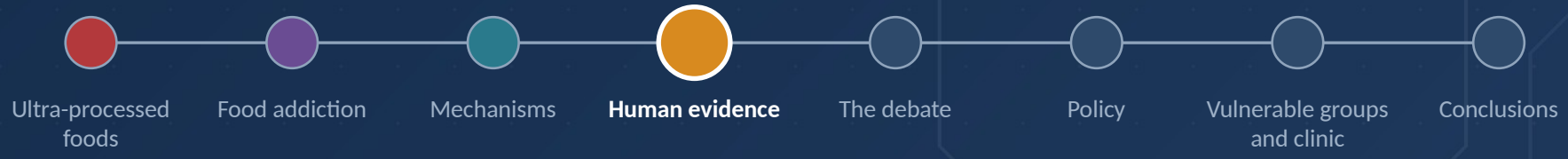
HYPOTHESIS-GENERATING

*Less craving after a drug does not prove addiction: GLP-1 agonists also change appetite, satiety and intake.*

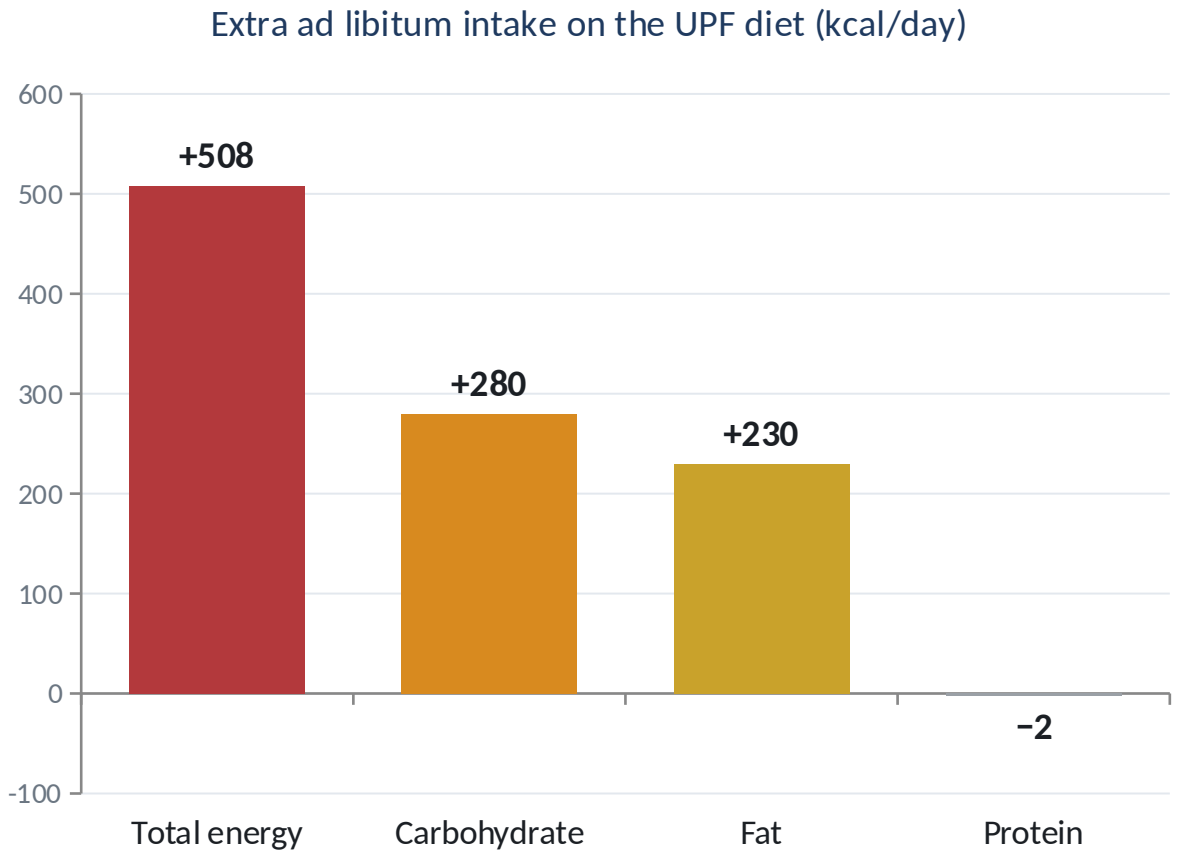
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# 04 Human Evidence

*What do trials and cohorts actually show?*



# Hall et al. 2019: the inpatient feeding trial




*Protein intake unchanged — consistent with protein leverage, but the trial was not designed to test it.*

**n = 20 adults · 4 weeks inpatient · crossover**
2 weeks UPF + 2 weeks minimally processed; diets matched for presented energy, macronutrients, sugar, sodium and fiber; eat as much as wanted

**+508**
kcal/day more on the UPF diet

**+0.9 kg**
vs −0.9 kg on the minimally processed diet (2 weeks)

**48 vs 31**
kcal per minute: faster eating on UPF

**Limits:** small, short, inpatient; diets also differed in non-beverage energy density, eating rate and texture. **No measure of addictive-like eating.**

*Source: Hall et al. 2019 (Cell Metabolism). Values are mean differences; error bars omitted.*

# Dicken et al. 2025: UPF within dietary guidelines

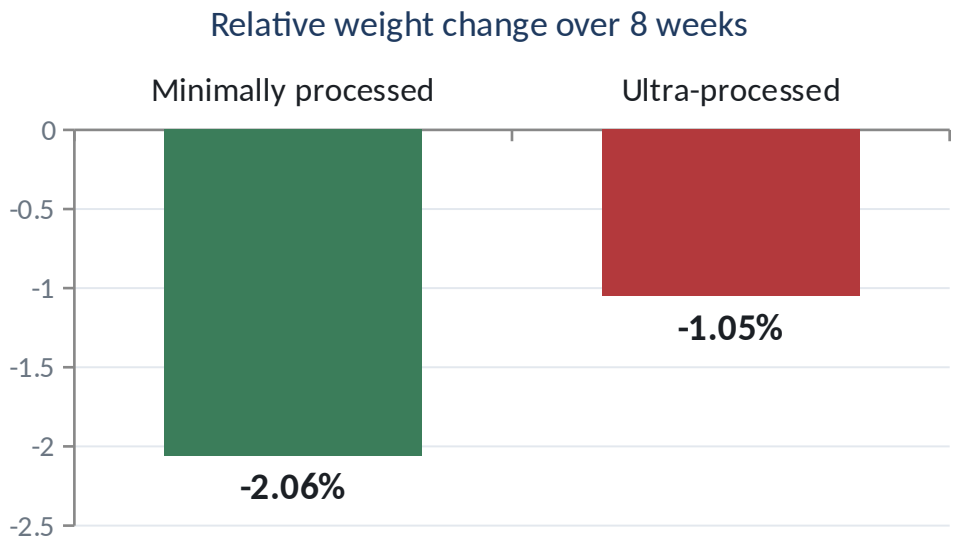
## Randomized crossover trial, England

55 randomized · 50 in intention-to-treat · BMI 25 to <40 · habitual UPF ≥50% of energy

**Diet A · 8 weeks** | **Washout · 4 weeks** | **Diet B · 8 weeks**

*random order*

**Both diets followed the UK Eatwell Guide,** ad libitum and delivered to participants: one mainly minimally processed, one mainly ultra-processed.




## Difference −1.01 percentage points

95% CI −1.87 to −0.14; p = 0.024. Fat mass also fell more on the minimally processed diet, and some craving measures moved the same way.

## Neither trial isolates "processing"

Diets were bundles: energy density, eating rate, sensory properties and palatability changed together. Of 106 national guidelines, most still give nutrient-based messages (Koios et al. 2022).

*Sources: Dicken et al. 2025 (Nature Medicine); Hall et al. 2019; Koios, Machado and Lacy-Nichols 2022.*

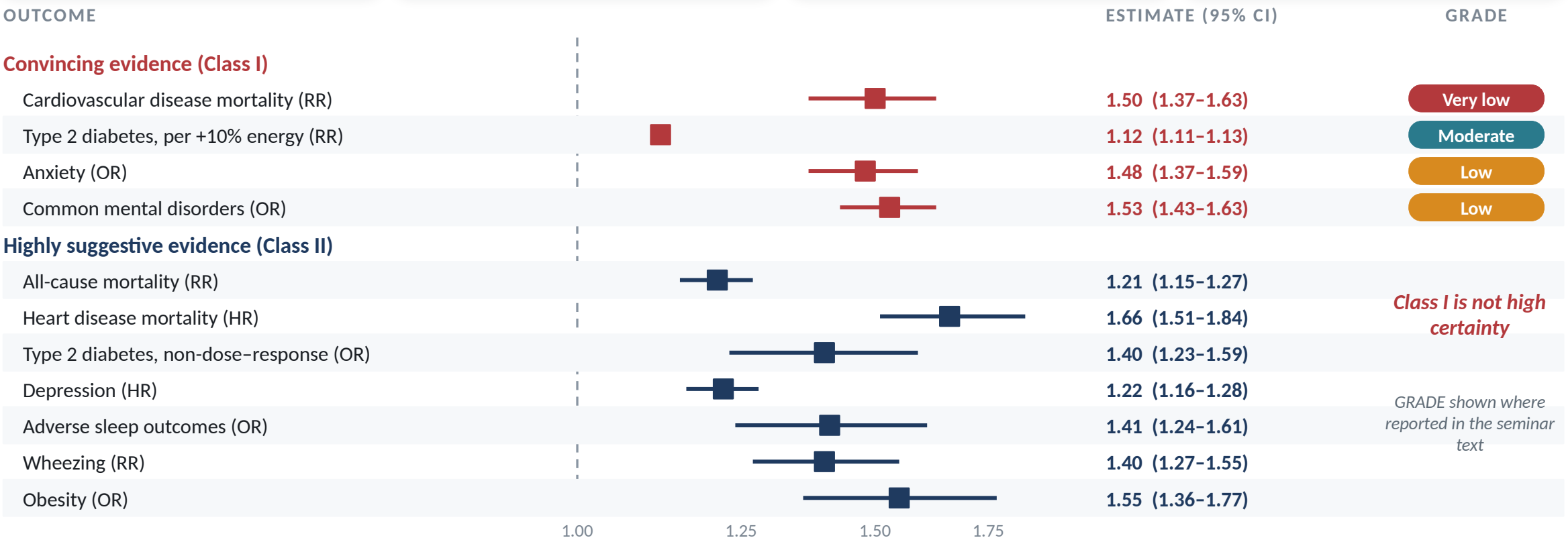
# Lane et al. 2024: umbrella review of 45 pooled analyses

**45**
pooled analyses from 14 meta-analyses

**9,888,373**
participations — not unique individuals

**32 / 45**
(71%) direct associations with harm

**0 · 4 · 22 · 19**
GRADE high · moderate · low · very low




*Source: Lane et al. 2024 (BMJ). Class = credibility of association; GRADE = certainty of evidence. Mainly observational data.*

# Large cohorts: consistent signals, heterogeneous outcomes

## NutriNet-Santé

France · n = 104,980

**HR 1.12** overall cancer per +10% UPF energy (1.06–1.18)

- Breast cancer HR 1.11 (1.02–1.22)
- Robust to diet-quality adjustment
- Residual confounding still possible

## EPIC

Europe · 25 cancer sites

**Mixed** association pattern differs by cancer site

- Processed and ultra-processed foods examined
- Not uniform across populations or sites
- UK Biobank and Moli-sani: similar signals

## TLGS

Tehran · median follow-up 7 years

**HR 1.28** prediabetes, top vs bottom tertile (1.07–1.52)

- n = 1,954; 766 prediabetes cases
- Per +10% energy: HR 1.12 (1.02–1.23)
- Type 2 diabetes (n = 2,457): no significant link



## The gap

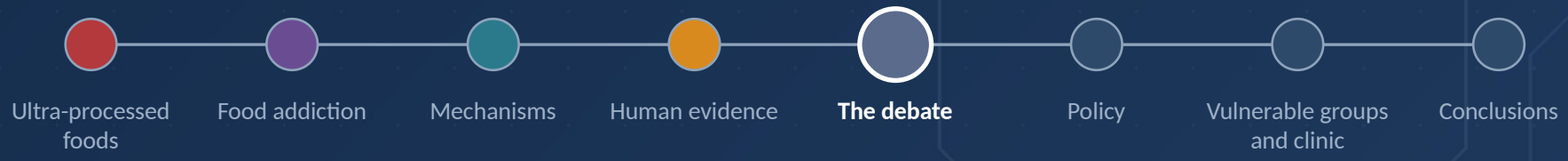
None of the trials, the umbrella review or these cohorts measured addictive-like eating or YFAS scores. They answer claim (a) — energy intake, weight, health — not claim (b).

*Sources: Fiolet et al. 2018; Kliemann et al. 2023; Moslehi et al. 2024; Lane et al. 2024; Schulte et al. 2015.*

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# 05 The Debate

*Is “addictive” the right word?*



# Two competing narratives

## The case for

**Gearhardt and DiFeliceantonio 2023**

Tobacco criteria (US Surgeon General, 1988) applied to highly processed foods:

**Compulsive use**

despite wanting to stop

**Psychoactive effects**

changes mood or mental state

**Reinforcement**

repeats consumption

**Strong cravings**

added later from addiction science

*No single active ingredient needed: as with tobacco, delivery speed and product design matter.*

VS

## The case against

**Ziauddeen, Fletcher, Hebebrand and colleagues**

1. **No addictive agent**

   Sugar? Fat? A combination? The agent is unspecified (Ziauddeen and Fletcher 2013).

2. **Heterogeneous imaging**

   Human findings are inconsistent, sometimes contradictory (Ziauddeen et al. 2012).

3. **Wrong term**

   “Eating addiction”: the behavior, not a substance, is addictive-like (Hebebrand et al. 2014).

*Sugar addiction: human evidence scant; animal data rely on intermittent access (Westwater et al. 2016).*

**Still live:** Fletcher vs Kenny (2018) and the 2021 Gearhardt vs Hebebrand debate — about construct validity, interpretation and consequences, not just data.

# Simpler explanations — and methodological limits

## Rule these out before invoking addiction

### Energy density

More energy per gram → more energy at the same volume (Rolls 2009)

### Protein leverage

Diluted protein → eat more to hit a protein target (Simpson and Raubenheimer 2005)

### Eating rate

Soft texture, less chewing → faster eating → more intake (Robinson et al. 2014)

### Restraint–binge cycle

Dietary restriction followed by overeating (Fairburn et al. 2003)

**Not mutually exclusive:** weights differ between people; addiction may fit a subgroup.

## Methodological limits

**Dietary assessment**

FFQs and recalls misclassify NOVA groups

**Residual confounding**

UPF intake tracks social and lifestyle factors

**Reverse causation**

Mental health ↔ UPF may run both ways

**Excess significance**

Small-study effects in some pooled analyses

**Self-report**

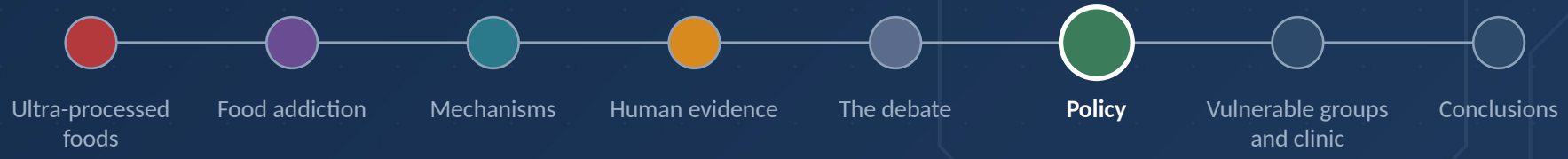
YFAS relies on what people say

*Sources: Rolls 2009; Simpson and Raubenheimer 2005; Robinson et al. 2014; Fairburn et al. 2003; LaFata et al. 2024; Lane et al. 2022, 2024.*

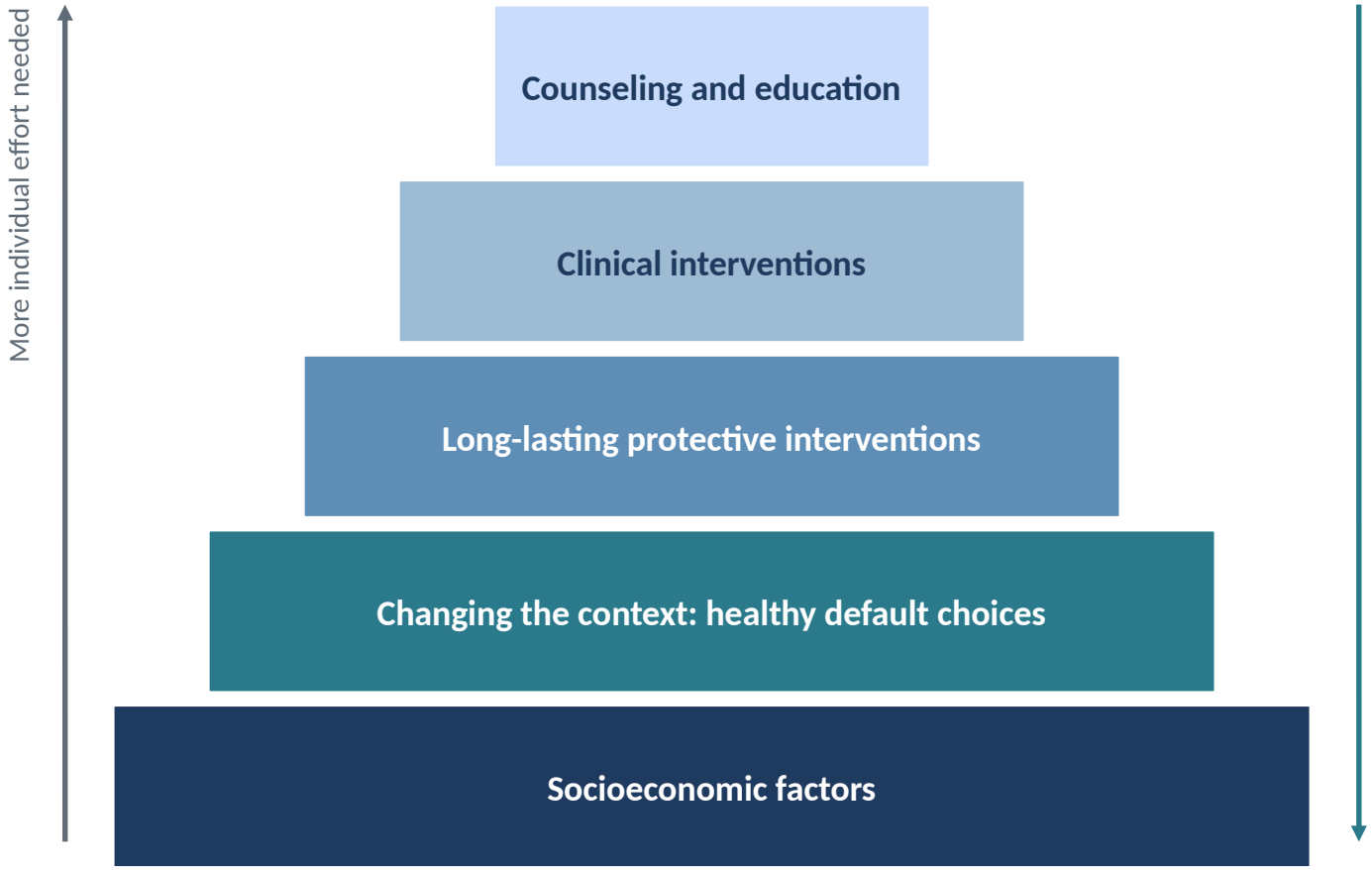
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# 06 Policy

*What can be done without settling the debate?*



# Why individual education is not enough




**Environment over willpower.** If products and settings drive intake beyond need, education alone shifts the burden onto individuals; environmental measures work without constant self-control.

**Equity.** Downstream campaigns may favor advantaged groups; structural and fiscal measures can narrow gaps. Mexico: largest drop in lowest-SES households.

**Read outcomes carefully.** Most evaluations measure purchases, reformulation or ad exposure — not intake or disease.

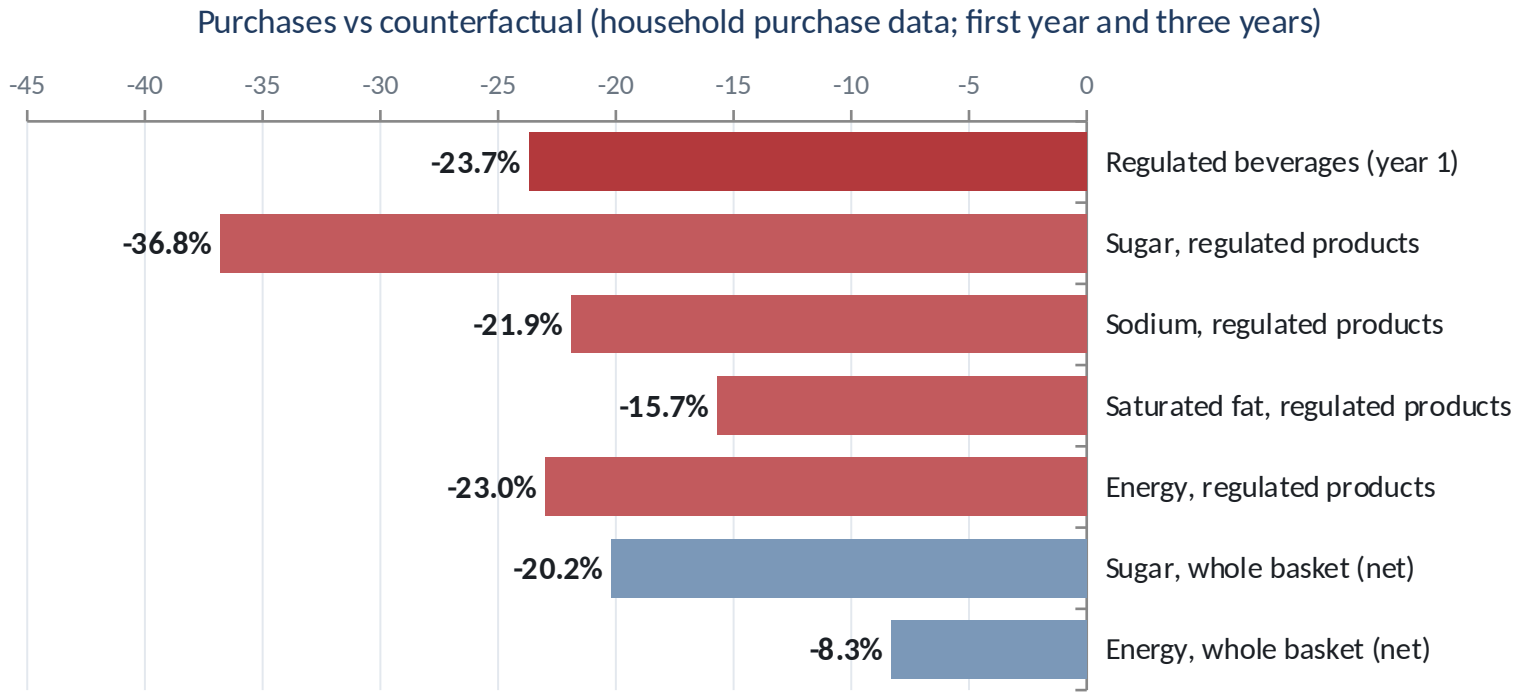
*Sources: Frieden 2010; Swinburn et al. 2011; Lorenc et al. 2013; Colchero et al. 2017.*

# Chile’s Law 20,606: warning labels as a policy package




**Three components, implemented in phases from 2016**

- Black octagon warning labels
- Marketing restrictions to children
- No sale of “high-in” products in schools



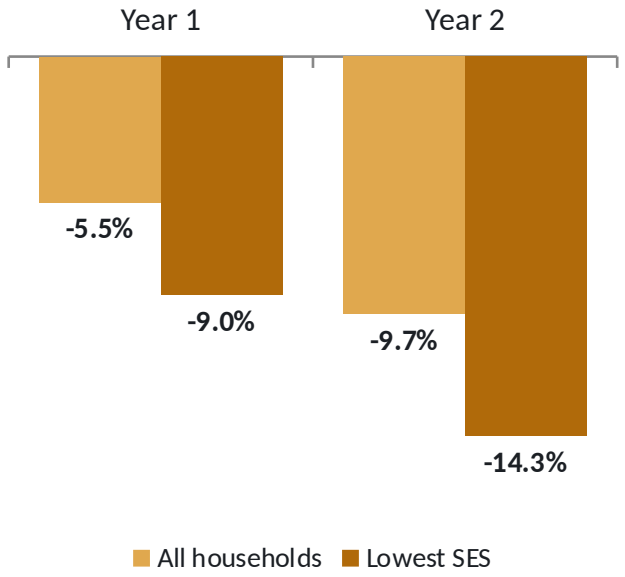

**Effect of the package, not of the label alone.** Purchases are not intake; changes were similar across socioeconomic groups; data cover only part of household purchases.

*Sources: Taillie et al. 2020 (first-year beverage purchases); Taillie et al. 2024 (three-year purchases).*

# Sugar-sweetened beverage taxes: flat vs tiered

## Mexico, 2014: flat tax

1 peso per liter on sugar-sweetened beverages




- Year 1: −6% on average, −12% by December; untaxed drinks +4%
- Two-year mean: −7.6%
- Equity is not automatic: regressive burden and substitution must be measured

## United Kingdom, 2018: tiered levy

Announced 2016 · implemented April 2018 · paid by producers

| | |
|---|---|
| **≥ 8 g sugar / 100 mL** | £0.24 per liter |
| **5 to < 8 g / 100 mL** | £0.18 per liter |
| **< 5 g / 100 mL** | exempt |

**15.4%** of drinks still above the lower threshold by February 2019

**−33.8 pp** drinks above 5 g/100 mL vs pre-policy trend — mainly via reformulation

**For Iran:** a tiered per-gram sugar tax merits modelling — not yet a recommendation. Needs local data on price elasticity, substitution, inflation and enforcement.

*Sources: Colchero et al. 2016, 2017; Scarborough et al. 2020. UK data not sales-weighted: purchases cannot be inferred.*

# Marketing, schools, guidelines — and the tobacco analogy

## Marketing to children

- **WHO 2010:** recommendations on marketing to children
- **UK:** “less healthy” TV ads before 9 p.m. and paid online ads restricted from 5 January 2026
- **Chile:** TV ad exposure −44% (preschoolers), −58% (adolescents)

## School food environment

- Limit sales of low-nutrition products; remove sugary drinks from vending machines
- Strengthen school meals built on fresh, minimally processed foods
- **Chile:** school sales ban was part of the package

## Processing in dietary guidelines

- **Brazil 2014 “golden rule”:** prefer natural or minimally processed foods and freshly made meals to UPF
- Followed to varying degrees by Uruguay, Peru, Ecuador, France, Belgium and Canada (106 guidelines reviewed)

## The tobacco analogy

- **1980–2001:** US tobacco companies owned major food companies; their products were 29% (fat–sodium) and 80% (carb–sodium) more likely to be hyper-palatable
- **Limits:** food is essential, UPF is heterogeneous, there is no single “nicotine”

*Sources: WHO 2010; UK Advertising Regulations 2024; Dillman Carpentier et al. 2020; Brazil MoH 2014; Koios et al. 2022; Fazzino et al. 2024; Gilmore et al. 2023.*

# Options for Iran: evidence, facilitators, barriers

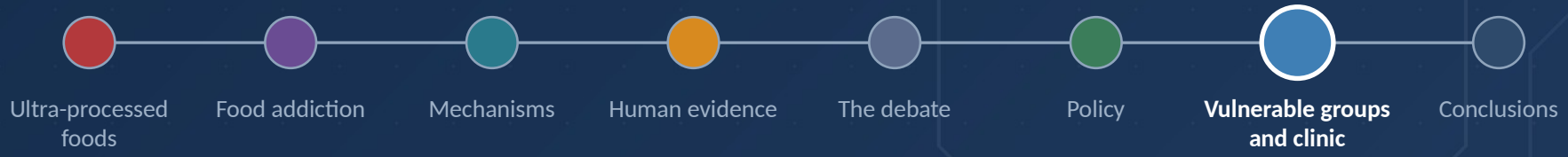
| Intervention | Example | Documented effect (intermediate) | Facilitator in Iran | Barrier in Iran |
|---|---|---|---|---|
| **Warning-label package** | Chile 2016 | Regulated products: sugar −36.8%, energy −23.0% | Existing nutrition-labeling infrastructure | Regulatory revision; inter-agency coordination; industry resistance |
| **Tiered sugar tax** | UK SDIL 2018 | Drinks above 5 g/100 mL: −33.8 pp (reformulation) | Existing tax system | Tax–FDA coordination; distribution and substitution |
| **Flat SSB tax** | Mexico 2014 | Purchases −7.6% (two-year mean); largest in lowest SES | Simple to implement | Shift to untaxed drinks; regressivity |
| **Marketing limits for children** | Chile; WHO 2010 | TV ad exposure −44% (preschool), −58% (adolescents) | Oversight of official media | Digital and social media hard to monitor |
| **School food environment** | Chile, Brazil | Part of packages; independent effect unclear | Ministry of Education canteen rules | Canteen economics; healthy alternatives; monitoring |
| **Processing in guidelines** | Brazil 2014 | Message framework shifts; present in many guidelines | Little new infrastructure needed | Public communication; behavioral effect unevaluated |
| **National UPF monitoring** | NHANES, NDNS | Enables trends, burden and policy evaluation | Add a module to existing surveys | Cost; standard NOVA coding; coder training |

*Feasibility columns are the author's qualitative assessment (infrastructure, cost, monitoring capacity, stakeholder resistance) — not a formal feasibility or stakeholder analysis. Highlighted row: a prerequisite for evaluating all the others.*

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# 07 Vulnerability and the Clinic

*Who is most at risk — and what should clinicians say?*



# Who is most vulnerable?

## Children and adolescents

**15%** all samples

**12%** community

**19%** overweight

*YFAS-defined food addiction (Yekaninejad et al. 2021)*

- Prefrontal control still maturing: reward–control imbalance (Casey et al. 2008)
- Heavy exposure to food marketing
- Food preferences form early but stay malleable (Birch 1999)

## Obesity and eating disorders

**11.1%** healthy weight

**24.9%** overweight or obesity

*Correlation, not identity: BMI is no proxy (Pursey et al. 2014)*

- BED: 92.4% met DSM-IV dependence criteria with “binge eating” in place of “substance” (n = 79; Cassin and von Ranson 2007)
- 12/12 with BED and 5/12 without met mild SUD criteria (qualitative; Curtis and Davis 2014)

## Mental health and income

**OR 1.44** depression

**OR 1.48** anxiety

*Mostly cross-sectional; possibly bidirectional (Lane et al. 2022)*

- US data: stronger link with food insecurity (LaFata et al. 2024)
- Iran: inflation → more cereals, less fruit; sanctions → most food groups fell, more so in low-income homes, bread rose
- Restrictive UPF policy needs affordable alternatives

*Sources: Yekaninejad et al. 2021; Casey et al. 2008; Birch 1999; Pursey et al. 2014; Cassin and von Ranson 2007; Curtis and Davis 2014; Lane et al. 2022; LaFata et al. 2024; Rahbarinejad et al. 2025; RezaeeDaryakenari et al. 2024.*

# Clinical implications: abstinence, moderation, stigma

## Abstain from "trigger foods"?

- Food itself cannot be given up; proposals target specific foods or features
- Peer-support models (e.g., Overeaters Anonymous) are not evidence-based treatment
- One 10–14-week UPF-abstinence pilot; rigorous RCTs still needed
- Restriction can maintain binge eating

## What already works

- CBT and interpersonal therapy: strong evidence in binge-eating disorder (Hilbert et al. 2019)
- Harm reduction and moderation are accepted goals in addiction care (Marlatt and Witkiewitz 2010)
- Open question: does the addiction model add clinical value?

## Stigma cuts both ways

+ Addiction framing reduced blame and stigma in an experiment (Latner et al. 2014)

– Belief in food addiction may prompt restriction → craving → disinhibited eating; still a hypothesis (Ruddock and Hardman 2017)

**Clinical bottom line:** no blanket advice to abstain from trigger foods; individualize after clinical assessment, and use addiction language without denying the person's agency.

*Sources: LaFata et al. 2024; Fairburn et al. 2003; Hilbert et al. 2019; Marlatt and Witkiewitz 2010; Hebebrand and Gearhardt 2021; Latner et al. 2014; Ruddock and Hardman 2017.*

# Limitations of this review and eight research gaps

## Limitations

- Narrative review: possible selection bias
- No predefined criteria; no uniform GRADE appraisal
- NIH four-arm trial (NCT05290064) completed but unpublished: data preliminary
- Iranian evidence fragmented and partly indirect
- Fast-moving field: needs periodic updating

**1** **Active factor**

Multi-arm factorial feeding trials

**2** **Construct validity**

YFAS + structured interviews + latent class analysis

**3** **Natural history**

Cohorts from childhood into adulthood

**4** **Withdrawal**

Energy-matched selective-elimination trials

**5** **Treatment**

RCTs vs CBT with long-term follow-up

**6** **UPF classification**

Inter-coder reliability; compare systems

**7** **Iranian intake data**

National survey processing module

PRIORITY FOR IRAN

**8** **Policy evaluation in Iran**

Interrupted time series of traffic-light labels

PRIORITY FOR IRAN

# Take-home messages

## 1 UPF and health

RELATIVELY STRONG

Trials and an umbrella review link UPF to higher intake, weight gain and 32 of 45 outcomes — though most observational GRADE is low.

## 2 Food addiction

UNRESOLVED

Not in DSM-5-TR or ICD-11; no identified agent; heterogeneous imaging. Unresolved, but not refuted.

## 3 Stay open and testable

LIMITED

Energy density, eating rate and protein leverage explain much; addiction may fit an undefined subgroup.

## 4 Act now

STRONG (INTERMEDIATE)

Labels, taxes, marketing and school limits shift purchases, recipes and ad exposure — without settling the debate.

## 5 Priorities for Iran

- Regular national monitoring of UPF in the household basket and diets
- Strengthen and evaluate the existing labeling system
- Protect the school food environment

*Priorities for evaluation — not recommendations based on Iranian evidence.*

**Answer to the guiding question:** no — evidence-based action on ultra-processed foods need not wait for the addiction debate to end.

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# Thank you

## *Questions and discussion*

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### KEY READING


- Hall KD et al. Cell Metab 2019 — inpatient UPF trial
- Dicken SJ et al. Nat Med 2025 — UPF within dietary guidelines
- Lane MM et al. BMJ 2024 — umbrella review
- Gearhardt AN, DiFeliceantonio AG. BMJ 2023 — UPF as addictive substances
- Ziauddeen H, Fletcher PC. Obes Rev 2013 — is food addiction valid?
- Taillie LS et al. PLOS Med 2020 — Chile's law

# Backup: arguments for and against food addiction

| Domain | Supportive view | Critical view |
|---|---|---|
| **Addictive agent** | Nutrient combinations, delivery speed and food matrix may raise reinforcing potential | No agent equivalent to a drug has been identified |
| **Neural evidence** | Overlap in reward circuits and dopamine; DRD2/OPRM1 markers; supra-additive fat + carb response | Human imaging heterogeneous, sometimes contradictory |
| **Withdrawal** | ProWS: self-reported symptoms after cutting down — preliminary evidence | Not separable from energy deficit or dietary restriction |
| **Terminology** | “Food addiction” highlights product properties and the food environment | “Eating addiction” is more accurate: the behavior is addictive-like |
| **Diagnostic status** | No diagnosis ≠ no phenomenon; gambling entered the classifications | Evidence insufficient; absent from DSM-5-TR and ICD-11 |
| **Policy implication** | Can support food-environment interventions | Premature as an independent basis for policy |
| **Stigma** | Can reduce blame and stigma (Latner et al. 2014) | Belief may prompt restriction and adverse eating (Ruddock and Hardman 2017) |

*Table 4 of the seminar. Sources: Gearhardt and Schulte 2021; Gearhardt and DiFeliceantonio 2023; Ziauddeen and Fletcher 2013; Ziauddeen et al. 2012; Hebebrand et al. 2014; Schulte et al. 2018.*

# Backup: key numbers at a glance

| | | | | | |
|---|---|---|---|---|---|
| **58%**<br>UPF share of energy, US adults<br>*Steele 2023* | **54% · 66%**<br>UK adults · adolescents<br>*Rauber 2020* | **13.4%**<br>Tehran adults, median (TLGS)<br>*Moslehi 2024* | **14% · 12%**<br>YFAS food addiction: adults · children<br>*Gearhardt 2023* | **11.1% · 24.9%**<br>healthy weight · overweight or obesity<br>*Pursey 2014* | **92.4%**<br>BED meeting dependence criteria (n = 79)<br>*Cassin 2007* |
| **+508**<br>kcal/day on the UPF diet<br>*Hall 2019* | **48 vs 31**<br>kcal/min eating rate, UPF vs minimal<br>*Hall 2019* | **−2.06% · −1.05%**<br>weight change, minimal vs UPF diet<br>*Dicken 2025* | **32 / 45**<br>outcomes with direct associations<br>*Lane 2024* | **0 · 4 · 22 · 19**<br>GRADE high · moderate · low · very low<br>*Lane 2024* | **HR 1.12**<br>cancer per +10% UPF energy<br>*Fiolet 2018* |
| **HR 1.28**<br>prediabetes, top vs bottom tertile<br>*Moslehi 2024* | **−23.7%**<br>regulated beverage purchases, year 1<br>*Taillie 2020* | **−36.8%**<br>sugar purchased, regulated products<br>*Taillie 2024* | **−7.6%**<br>taxed beverages, two-year mean<br>*Colchero 2017* | **−33.8 pp**<br>UK drinks above 5 g/100 mL<br>*Scarborough 2020* | **−44% · −58%**<br>TV ad exposure, preschool · teens<br>*Dillman Carpentier 2020* |

*Descriptive figures from different designs and populations: compare with care.*